

ParaCodex: A Profiling-Guided Autonomous Coding Agent for Reliable Parallel Code Generation and Translation



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Keywords: autonomous coding agents, tool-using LLMs, OpenMP GPU offload, parallel code generation, code translation, profiling-guided optimization, program transformation, performance feedback

Abstract:

Parallel programming is central to HPC and AI, but producing code that is correct and fast remains challenging, especially for OpenMP GPU offload, where data movement and tuning dominate. Autonomous coding agents can compile, test, and profile on target hardware, but outputs are brittle without domain scaffolding.

We present ParaCodex, an HPC-engineer workflow that turns a Codex-based agent into an autonomous OpenMP GPU offload system using staged hotspot analysis, explicit data planning, correctness gating, and profiling-guided refinement. We evaluate translation from serial CPU kernels to OpenMP GPU offload kernels on HeCBench, Rodinia, and NAS. After excluding five kernels, ParaCodex succeeded on all 31 valid kernels. The generated kernels improved GPU time over reference OpenMP implementations in 25/31 cases, achieving geometric-mean speedups of $3\times$ on HeCBench and $5\times$ on Rodinia, and outperforming a zero-shot Codex baseline on all suites. We also evaluate CUDA→OpenMP offload translation on ParEval, where ParaCodex maintains high compilation and validation rates in code-only and end-to-end settings.

ParaCodex anonymous repository: <https://doi.org/10.5281/zenodo.18148256> (<https://doi.org/10.5281/zenodo.18148256>)

Paper Type: Long

Research Area: AI/LLM Agents

Research Area Keywords: Autonomous agents, LLM agents, tool use, function calling, planning in agents, environment interaction, agent memory, agent evaluation, safety and alignment for agents, grounded agents

Contribution Types: NLP engineering experiment, Publicly available software and/or pre-trained models, Data analysis

Languages Studied: C/C++, OpenMP, CUDA

Reassignment Request Area Chair: This is not a resubmission

Reassignment Request Reviewers: This is not a resubmission

Justification For Not Keeping Action Editor Or Reviewers: N/A

Software:  zip (/attachment?id=xkW1rafZ8v&name=software)

A1 Limitations Section: This paper has a limitations section.

A2 Potential Risks: N/A

B Use Or Create Scientific Artifacts: Yes

B4 Data Contains Personally Identifying Info Or Offensive Content: N/A

B6 Statistics For Data: Yes

B6 Elaboration: §4.1 (Table 1) and §4.3.

C Computational Experiments: Yes

C2 Experimental Setup And Hyperparameters: Yes

C2 Elaboration: §4.1 and §3.2.

C3 Descriptive Statistics: Yes

C3 Elaboration: §4.1 and §4.3.

D Human Subjects Including Annotators: No

D1 Instructions Given To Participants: N/A

D2 Recruitment And Payment: N/A

D3 Data Consent: N/A

D4 Ethics Review Board Approval: N/A

E Ai Assistants In Research Or Writing: Yes

E1 Information About Use Of Ai Assistants: Yes

E1 Elaboration: §3 and §4.1.

Author Submission Checklist: yes

TLDR:  ParaCodex is a profiling-guided Codex agent that reliably generates and translates parallel code using staged analysis, explicit data planning, correctness gates, and profiling-in-the-loop optimization.

Justification For Author Changes:  N/A

Preprint:  no

Preprint Status:  We are considering releasing a non-anonymous preprint in the next two months (i.e., during the reviewing process).

Existing Preprints:  N/A

Preferred Venue:  ACL

Consent To Share Data:  yes

Consent To Share Submission Details:  On behalf of all authors, we agree to the terms above to share our submission details.

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Submission Number: 6332

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Official Review of Submission6332 by Reviewer 4Sha

Official Review by Reviewer 4Sha 📅 14 Feb 2026 at 02:18 (modified: 15 Feb 2026 at 14:54)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer 4Sha, Authors

📄 Revisions (/revisions?id=duOaFmdIqz)

Paper Summary:

This paper presents PARACODEX, an autonomous coding workflow that turns a large language model into a practical OpenMP GPU offloading system by embedding HPC engineering practices into an agentic pipeline.

The authors argue that while LLMs can generate plausible parallel code, they often fail in real HPC settings due to issues such as incorrect data movement, unsafe parallelization, and performance regressions.

Compared to expert-provided OpenMP implementations, PARACODEX improves GPU execution time in 25 out of 31 kernels, with geometric-mean speedups of 3× on HeCBench, 5× on Rodinia, and modest gains on the highly optimized NAS suite.

It also outperforms a zero-shot Codex baseline across all benchmark suites.

Overall, the paper demonstrates that combining LLMs with structured domain scaffolding, tool use, and performance feedback can transform brittle code generation into a robust, performance-driven autonomous HPC programming system.

Summary Of Strengths:

- This ParaCodex framework explicitly introducing the profiling feedback and regression rollback to optimize runtime performance.
- This paper provides extensive code and detailed implementation.
- Different from other code generation framework, this ParaCode can generalize to CUDA-to-OpenMP translation, besides, PARACODEX can implement pseudo-offload cases and enforce device-side work.

Summary Of Weaknesses:

- This topic is more related to software engineering with LLMs; therefore, ACL might not be a suitable venue. Software engineering venues would be more appropriate.
- The experiments focus more on feasibility and generalization, which are more relevant to software engineering rather than the language field. A rigorous and comprehensive quantitative evaluation is required.
- The experiments are mainly conducted in a single-GPU setting, lacking demonstration in multi-GPU frameworks.

Comments Suggestions And Typos:

Please refer to the weakness section.

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 4 = Useful: I would recommend the new software to other researchers or developers for their ongoing work.

Knowledge Of Or Educated Guess At Author Identity: No

Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

Knowledge Of Paper Source: N/A, I do not know anything about the paper from outside sources

Impact Of Knowledge Of Paper: N/A, I do not know anything about the paper from outside sources

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Publication Ethics Policy Compliance: I did not use any generative AI tools for this review

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Official Review of Submission6332 by Reviewer y1k1

Official Review by Reviewer y1k1 📅 11 Feb 2026 at 03:50 (modified: 15 Feb 2026 at 14:54)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer y1k1, Authors

📄 Revisions (/revisions?id=srBW4mDeDR)

Paper Summary:

This paper presents PARACODEX, an autonomous coding agent for serial-to-OpenMP GPU offload translation and CUDA-to-OpenMP migration. Rather than relying on one-shot code generation, the system employs a staged, artifact-driven workflow that mirrors the iterative practice of an HPC engineer: (1) hotspot analysis using a loop taxonomy and estimated computational weight; (2) explicit data planning to reason about data residency and transfer strategies prior to code transformation; (3) correctness gating via compilation and checksum-based numerical validation; and (4) profile-guided refinement using NVIDIA Nsight Systems feedback, with a strict revert-on-regression policy. Evaluated on HeCBench, Rodinia, NAS, and ParEval, PARACODEX produces correct GPU offload implementations for all 31 valid kernels, improves GPU time in 25/31 cases, achieves geometric-mean speedups of 3× and 5.1× on HeCBench and Rodinia respectively, and approaches expert-level performance on the NAS benchmark suite.

Summary Of Strengths:

- 1/ The paper addresses a realistic and important challenge: generating OpenMP GPU offload code that is both correct and efficient. The motivation is clearly articulated, and the failure modes of naive LLM parallelization (e.g., data thrashing) are also clearly described.
- 2/ The central contribution ie., artifact-driven agentic workflow is well-defined and consistently implemented.
- 3/ Unlike prior LLM-based translation pipelines that focus primarily on compilation and test repair, PARACODEX explicitly incorporates profiling feedback and regression rollback to optimize runtime performance.
- 4/ The system is evaluated across multiple benchmark suites (HeCBench, Rodinia, NAS, ParEval). The comparison against a zero-shot Codex baseline using identical constraints isolates the impact of the workflow.
- 5/ The reasoning trace comparison in Sec. 4.5 strengthens the claim that structured workflows meaningfully alter agent behavior.

Summary Of Weaknesses:

The weaknesses can be grouped into three: (1) evaluation scope, (2) measurement and validation rigor, and (3) system robustness.

Most are explicitly acknowledged by the authors and primarily limit generalizability and experimental depth rather than the core method contribution. I feel these constraints do not undermine the paper's central claims.

1/ Evaluation is restricted to a single consumer-grade RTX 4060 Laptop GPU, leaving performance dynamics on enterprise-grade clusters or multi-GPU environments unverified.

2/ GPU times are averaged over three runs without variance or significance reporting. Large speedups are convincing, but smaller gains (e.g., near-parity results on NAS) would benefit from stronger statistical support.

3/ GPU offload bypass occurred in some kernels. While detected and excluded, this shows a robustness gap in enforcing device execution constraints.

4/ The token budget per kernel is substantial. Although justified for reproducibility, scalability concerns remain.

Comments Suggestions And Typos:

NA

Confidence: 3 = Pretty sure, but there's a chance I missed something. Although I have a good feel for this area in general, I did not carefully check the paper's details, e.g., the math or experimental design.

Soundness: 3 = Acceptable: This study provides sufficient support for its main claims. Some minor points may need extra support or details.

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3 = Findings: I think this paper could be accepted to the Findings of the ACL.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 3 = They could reproduce the results with some difficulty. The settings of parameters are underspecified or subjectively determined, and/or the training/evaluation data are not widely available.

Datasets: 1 = No usable datasets submitted.

Software: 3 = Potentially useful: Someone might find the new software useful for their work.

Knowledge Of Or Educated Guess At Author Identity: No

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Official Review of Submission6332 by Reviewer n5bM

Official Review by Reviewer n5bM 📅 01 Feb 2026 at 05:15 (modified: 15 Feb 2026 at 14:54)

👁 Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer n5bM, Authors

📄 Revisions (/revisions?id=ISiHyrp5eC)

Paper Summary:

This paper introduces an end-to-end approach to parallel programming in High-Performance Computing (HPC) using a tool-driven autonomous Large Language Model (LLM) agent called PARACODEX. The agent aims to bridge the gap between automatic code generation and the necessity of efficient data movement and tuning in OpenMP GPU offload. The paper presents PARACODEX as a solution that integrates several methods include hotspot analysis, explicit data planning, correctness gating and profiling-guided refinement, to enhance the accuracy and robustness of parallel code generation for GPUs. The authors provide detailed evaluations of PARACODEX using multiple benchmark suites, such as HeCBench, Rodinia, and NAS, which cover a wide range of parallel kernels. The results demonstrate that PARACODEX succeeded in most of the tested kernels and achieves significant improvements in GPU time, with geometric mean speedups of 3× on HeCBench, 5× on Rodinia, compared to the existing reference OpenMP implementations. In comparison to a zero-shot Codex baseline, PARACODEX shows not only better performance but also a more reliable process for ensuring correctness and performance optimization.

Summary Of Strengths:

1. Systematization of HPC Engineering Practices: The paper effectively systematizes HPC engineering practices into a reusable agentic pattern. By integrating multiple stages, including hotspot analysis, loop taxonomy, data residency transfer strategy selection and profile-guided tuning.
2. Comprehensive Evaluation and Benchmarking: The paper presents a thorough and unified evaluation protocol for serial→OpenMP agentic parallelization across three benchmark suites. The agent not only assess robustness but also quantify improvements in GPU-time performance relative to expert implementations over baseline. Such experiments solidifies PARACODEX's ability to consistently improve parallel code performance.
3. Generalization and Cross-Platform Migration The paper showcases ability to generalize to CUDA→OpenMP translation, demonstrating its flexibility across different scenarios. With bypass-detection analysis, PARACODEX can surfaces pseudo-offload cases and enforce device-side work.

Summary Of Weaknesses:

1. Limited Experimental Setup with a Single Hardware Configuration: The experiments are conducted using a single hardware setup, specifically an NVIDIA RTX 4060 GPU and an i9-13905H CPU. While this setup provides valuable insights, it may not fully demonstrate the robustness and generalizability of the proposed method across different CPU architectures. Optimizations performed on this specific CPU may not translate well to other systems with different CPU designs.
2. Limited Number of Optimized Kernels: The number of kernels optimized in the experiments is relatively small. While the proposed method demonstrates strong performance on the selected kernels, its performance on more complex or irregular kernels is not yet fully explored. The method may perform well on simpler cases, but its effectiveness in handling more challenging and diverse kernels remains uncertain.
3. Single Baseline Model Used: The study only compares PARACODEX against a single baseline, gpt-5.1-codex-mini. While this model provides useful insights, other models, such as Gemini, may offer comparable or even better performance. Including additional baseline models in the evaluation could provide a more comprehensive understanding of PARACODEX's relative performance.
4. Lack of Comprehensive Time Analysis: While the paper provides time analysis for specific kernels, it lacks a complete time analysis of the entire PARACODEX workflow. The time cost of iterative refinement and its trade-off with performance gains are not fully explored, especially for complex kernels.

Comments Suggestions And Typos:

1. Hardware Diversity The evaluation is conducted on a single hardware configuration. Adding experiments on different CPU architectures or GPU platforms would help demonstrate the robustness and generalizability of the proposed approach.
2. Kernel Coverage The number of optimized kernels is relatively limited. Evaluating more complex kernels would provide a clearer understanding of how well the method scales to challenging real-world workloads.
3. Baseline Models The comparison is limited to a single baseline model (gpt-5.1-codex-mini). Including

additional strong models (e.g., Gemini) would strengthen the comparative analysis. 4. End-to-End Time Analysis
While per-kernel timing is reported, the paper does not provide end-to-end runtime for the full PARACODEX workflow. Reporting total optimization time would help assess practical efficiency.

Confidence: 4 = Quite sure. I tried to check the important points carefully. It's unlikely, though conceivable, that I missed something that should affect my ratings.

Soundness: 3.5

Excitement: 3 = Interesting: I might mention some points of this paper to others and/or attend its presentation in a conference if there's time.

Overall Assessment: 3.5 = Borderline Conference

Limitations And Societal Impact:

The authors clearly discuss several limitations of their work, including evaluation on a single consumer-grade GPU without locked clocks, potential gaps in correctness validation for subtle numerical or non-deterministic issues, GPU offload bypass failure modes where CPU fallback code violates parallelization intent, and the need for stronger statistical validation of modest speedups.

Ethical Concerns:

There are no concerns with this submission

Needs Ethics Review: No

Reproducibility: 4 = They could mostly reproduce the results, but there may be some variation because of sample variance or minor variations in their interpretation of the protocol or method.

Datasets: 1 = No usable datasets submitted.

Software: 3 = Potentially useful: Someone might find the new software useful for their work.

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